CHAPTER 6

The New Members: How Computers Are Different from Printing Presses

It's hardly news that we are living in the midst of an unprecedented information revolution. But what kind of revolution is it exactly? In recent years we have been inundated with so many groundbreaking inventions that it is difficult to determine what is driving this revolution. Is it the internet? Smartphones? Social media? Blockchain? Algorithms? AI?

So before exploring the long-term implications of the current information revolution, let's remind ourselves of its foundations. The seed of the current revolution is the computer. Everything else – from the internet to AI – is a by-product. The computer was born in the 1940s as a bulky electronic machine that could make mathematical calculations, but it has evolved at breakneck speed, taking on novel forms and developing awesome new capabilities. The rapid evolution of computers has made it difficult to define what they are and what they do. Humans have repeatedly claimed that certain things would forever remain out of reach for computers – be it playing chess, driving a car or composing poetry – but 'forever' turned out to be a handful of years.

We will discuss the exact relations between the terms 'computer', 'algorithm' and 'AI' toward the end of this chapter, after we first gain a better grasp of the history of computers. For the moment it is enough to say that in essence a computer is a machine that can potentially do two remarkable things: it can make decisions by itself, and it can create new ideas by itself. While the earliest computers could hardly accomplish such things, the potential was already there, plainly seen by both computer